Comprehensive HR Data Analysis

Objective: Analyze an integrated HR database to understand the factors affecting performance and job satisfaction, identify the reasons employees leave the company, and evaluate the efficiency of recruitment and promotion processes, in order to provide strategic recommendations that strengthen workplace stability.

Description: A data project focused on merging and analyzing 7 different files (sheets) containing data on 10,000 employees. It includes: employee demographic and job details, the structure of 8 departments, 33 job titles, geographic location data, performance and engagement evaluation records, 2,653 promotion records and their impact on salary, and recruitment records covering cost, duration, and hiring sources.

Tools & Design: Designing interactive dashboards to visualize the data, along with data modeling diagrams to link the seven tables together.

Technologies to Use: Data cleaning and processing tools (such as Python/Pandas or Power Query), databases (SQL) for joining and extracting tables, and visualization tools (Power BI or Tableau).

Team Members & Responsibilities

Nahla Tarek Sayed Sayed: Analyze employee data (age, gender, salary, experience), as well as departments and geographic locations and their impact on the company's demographic distribution.

Eman Reda Tayel: Analyze performance evaluation records (10,000 records), job satisfaction scores, goal achievement rates, and employee engagement rates.

Roqaya Ahmed El-Sayed: Analyze promotion data (2,653 records), compare job levels before and after promotion, and calculate salary increase percentages and their impact on employee retention.

Ahmed Qarni Rabie: Analyze recruitment data, cost per hire, time to hire, candidate evaluations, and identify the most effective recruitment sources.

Week 1: [Data Exploration and Cleaning]

- Explore the data of the seven tables to identify missing values and outliers (with the participation of all team members).
- Clean the Employees, Departments, and Geographic Locations sheets (Nahla).
- Clean the Performance Evaluation, Promotions, and Recruitment sheets (Eman, Roqaya, Ahmed).

Deliverables:

- Clean, error-free data tables, with a preliminary report on data quality and status.

Week 2: [Data Modeling and Descriptive Analysis]

- Build a data relationship model to correctly link the seven tables together (all team members).
- Perform descriptive analysis and extract basic statistics on salaries and years of experience (Nahla and Eman).
- Analyze the reasons for employee departure based on the available records (Roqaya and Ahmed).

Deliverables:

- A connected Relational Data Model, and a report of preliminary descriptive statistics on employees.

Week 3: [Advanced Analytics]

- Prepare advanced analyses to answer business questions, such as the impact of promotions on job satisfaction and performance (Eman and Roqaya).
- Measure recruitment efficiency (cost return and time to hire) compared to the employee's performance after hiring (Ahmed).
- Analyze differences between the various departments and geographic locations (Nahla).

Deliverables:

- Individual analytical report drafts covering every angle of the data.

Week 4: [Integration and Final Presentation]

- Merge all analyses into one or more dashboards to make patterns and results easier to present (all team members).
- Draft the final recommendations and review the project presentation (all team members).

Deliverables:

- A final interactive dashboard (Final Dashboard) and a presentation (Presentation) showcasing the insights and recommendations resulting from the data analysis for management.